

## > Problem Statement

↳ 7 cells hidden

### ✓ Please read the instructions carefully before starting the project.

This is a commented Python Notebook file in which all the instructions and tasks to be performed are mentioned.

- Blanks '' ***are provided in the notebook that needs to be filled with an appropriate code to get the correct result. With every*** ' blank, there is a comment that briefly describes what needs to be filled in the blank space.
- Identify the task to be performed correctly, and only then proceed to write the required code.
- Please run the codes in a sequential manner from the beginning to avoid any unnecessary errors.
- Add the results/observations (wherever mentioned) derived from the analysis in the presentation and submit the same. Any mathematical or computational details which are a graded part of the project can be included in the Appendix section of the presentation.

**Note:** If the free-tier GPU of Google Colab is not accessible (due to unavailability or exhaustion of daily limit or other reasons), the following steps can be taken:

1. Wait for 12-24 hours until the GPU is accessible again or the daily usage limits are reset.
2. Switch to a different Google account and resume working on the project from there.
3. Try using the CPU runtime:
  - To use the CPU runtime, click on *Runtime* => *Change runtime type* => *CPU* => *Save*
  - One can also click on the *Continue without GPU* option to switch to a CPU runtime (kindly refer to the snapshot below)
  - The instructions for running the code on the CPU are provided in the respective sections of the notebook.

## Cannot connect to GPU backend

You cannot currently connect to a GPU due to usage limits in Colab. [Learn more](#)

To get more access to GPUs, consider purchasing Colab compute units with [Pay As You Go](#).

Close

Connect without GPU

## Installing and Importing Necessary Libraries and Dependencies

```
# Installation for GPU llama-cpp-python
# uncomment and run the following code in case GPU is being used
!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-py

# Installation for CPU llama-cpp-python
# uncomment and run the following code in case GPU is not being used
# !CMAKE_ARGS="-DLLAMA_CUBLAS=off" FORCE_CMAKE=1 pip install llama-cpp-py
```

```

1.8/1.8 MB 247.6 MB/s eta 0s
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
45.5/45.5 kB 321.1 MB/s eta 0s
16.7/16.7 MB 352.8 MB/s eta 0s
44.6/44.6 kB 288.4 MB/s eta 0s
Building wheel for llama-cpp-python (pyproject.toml) ... done
ERROR: pip's dependency resolver does not currently take into account
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-ex
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-p
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-s
grpcio-status 1.71.2 requires protobuf<6.0dev,>=5.26.1, but you have p
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.5.0 which
google-adk 1.29.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you l
google-adk 1.29.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you l
```

```
#pip install llama-cpp-python==0.1.85
```

```
#pip install langchain-text-splitters
```

```
#pip install sentence-transformers
```

```
#pip install faiss-cpu
```

```
#pip install pypdf
```

```
#pip install transformers
```

```
#pip install huggingface_hub
```

```
#pip install accelerate
```

```
#pip install ctransformers
```

```
#pip install tqdm
```

```
#pip install bitsandbytes
```

```
#pip install langchain
```

```
#pip install langchain-community
```

```
#pip install einops
```

```
#pip install protobuf==5.29.5
```

```
#pip install protobuf==6.31.1
```

**Note:**

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in *this notebook*.

## ◆ Gemini

```
# For installing the libraries & downloading models from HF Hub
!pip install huggingface_hub==0.23.2 pandas==1.5.3 tiktoken==0.6.0 pymupdf==1.25.1
pip install huggingface_hub==0.23.2 pandas==1.5.3 tiktoken==0.6.0 pymupdf==1.25.1
```

```
Installing build dependencies ... canceled
ERROR: Operation cancelled by user
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/base
    status = run_func(*args)
    ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/req
    return func(self, options, args)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/command
    requirement_set = resolver.resolve(
    ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    result = self._result = resolver.resolve(
    ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib
    state = resolution.resolve(requirements, max_rounds=max_rounds)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib
    self._add_to_criteria(self.state.criteria, r, parent=None)
  File "/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib
    if not criterion.candidates:
    ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib
    return bool(self._sequence)
    ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    return any(self)
    ^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    return (c for c in iterator if id(c) not in self.incompatible_id
```

```

            ^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    candidate = func()
            ^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    base: Optional[BaseCandidate] = self._make_base_candidate_from_li
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    self._link_candidate_cache[link] = LinkCandidate(
            ^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    super().__init__(
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    self.dist = self._prepare()
            ^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    dist = self._prepare_distribution()
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/resolut
    return preparer.prepare_linked_requirement(self._ireq, parallel_b
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/operati
    return self._prepare_linked_requirement(req, parallel_builds)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/operati
    dist = _get_prepared_distribution(
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/operati
    abstract_dist.prepare_distribution_metadata(
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/distrib
    self._prepare_build_backend(finder)
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/distrib
    self.req.build_env.install_requirements(
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/build_e
    self._install_requirements(
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/build_e
    call_subprocess(
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/utils/s
    line: str = proc.stdout.readline()
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/usr/local/bin/pip3", line 10, in <module>
    sys.exit(main())
            ^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/mai
    return command.main(cmd_args)
            ^^^^^^^^^^^^^^^^^^^^^

```

```

File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/bas
return self._main(args)
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/bas
return run(options, args)
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/bas
logger.debug("Exception information:", exc_info=True)
File "/usr/lib/python3.12/logging/__init__.py", line 1527, in debug
self._log(DEBUG, msg, args, **kwargs)
File "/usr/lib/python3.12/logging/__init__.py", line 1684, in _log
self.handle(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1700, in handl
self.callHandlers(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1762, in callH
hdlr.handle(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1028, in handl
self.emit(record)
File "/usr/lib/python3.12/logging/handlers.py", line 75, in emit
logging.FileHandler.emit(self, record)
File "/usr/lib/python3.12/logging/__init__.py", line 1280, in emit
StreamHandler.emit(self, record)
File "/usr/lib/python3.12/logging/__init__.py", line 1160, in emit
msg = self.format(record)
File "/usr/lib/python3.12/logging/__init__.py", line 999, in format
return fmt.format(record)
File "/usr/local/lib/python3.12/dist-packages/pip/_internal/utils/l
formatted = super().format(record)
File "/usr/lib/python3.12/logging/__init__.py", line 711, in format
record.exc_text = self.formatException(record.exc_info)
File "/usr/lib/python3.12/logging/__init__.py", line 661, in format
traceback.print_exception(ei[0], ei[1], tb, None, sio)
File "/usr/lib/python3.12/traceback.py", line 124, in print_excepti
te = TracebackException(type(value), value, tb, limit=limit, comp
File "/usr/lib/python3.12/traceback.py", line 733, in __init__
self.stack = StackSummary._extract_from_extended_frame_gen(
File "/usr/lib/python3.12/traceback.py", line 418, in _extract_from
for f, (lineno, end_lineno, colno, end_colno) in frame_gen:
File "/usr/lib/python3.12/traceback.py", line 355, in _walk_tb_with
positions = _get_code_position(tb.tb_frame.f_code, tb.tb_lasti)
File "/usr/lib/python3.12/traceback.py", line 369, in _get_code_pos
return next(itertools.islice(positions.gen, instruction index //

```

```
KeyboardInterrupt
^C
```

```
-----
AttributeError                                Traceback (most recent call
last)
/usr/lib/python3.12/pathlib.py in __str__(self)
    440         try:
--> 441             return self._str
    442         except AttributeError:
```

**AttributeError:** 'PosixPath' object has no attribute '\_str'

During handling of the above exception, another exception occurred:

```
KeyboardInterrupt                                Traceback (most recent call
last)
-----
17 frames
/usr/lib/python3.12/pathlib.py in root(self)
    558         return self._drv
    559
--> 560     @property
    561     def root(self):
    562         """The root of the path, if any."""
```

### Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in **this notebook**.

```
#Libraries for downloading and loading the llm
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

- ✓ Question Answering using LLM
- ✓ Downloading and Loading the model

```
model_name_or_path = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"
model_basename = "mistral-7b-instruct-v0.2.Q6_K.gguf"
```

```
model_path = hf_hub_download(
    repo_id= "TheBloke/Mistral-7B-Instruct-v0.2-GGUF", #Complete the code
    filename= "mistral-7b-instruct-v0.2.Q6_K.gguf" #Complete the code
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your setti
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to a
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthent
```

```
#uncomment the below snippet of code if the runtime is connected to GL
llm = Llama(
    model_path=model_path,
    n_ctx=2300,
    n_gpu_layers=38,
    n_batch=512
)
```

```
AVX = 1 | AVX2 = 1 | AVX512 = 0 | AVX512_VBMI = 0 | AVX512_VNNI = 0 |
```

```
#uncomment the below snippet of code if the runtime is connected to CI
#llm = Llama(
#    model_path=model_path,
#    n_ctx=1024,
#    n_cores=-2
#)
```

## ✓ Response

```
def response(query,max_tokens=128,temperature=0,top_p=0.95,top_k=50):
    model_output = llm(
        prompt=query,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k
    )

    return model_output['choices'][0]['text']
```

```
response("What treatment options are available for managing hypertens:
```

## ✓ Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = "What is the protocol for managing sepsis in a critical c
response(user_input)
```

```
'\n\nSepsis is a life-threatening condition that can arise from an inf
ection, and it requires prompt recognition and aggressive management i
n a critical care unit. The following are general steps for managing s
epsis in a critical care unit:\n\n1. Early recognition: Recognize the
signs and symptoms of sepsis early and initiate treatment as soon as p
ossible. Sepsis can present with various clinical features, including
```

## ✓ Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can
response(user_input_2) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nAppendicitis is a medical condition characterized by inflammation
of the appendix, a small tube-shaped organ located in the lower right
side of the abdomen. The symptoms of appendicitis can vary from person
to person, but some common signs include:\n\n1. Abdominal pain: The pa
in may start as a mild discomfort around the navel or in the lower rig
ht abdomen, which then gradually moves to the right lower quadrant and
becomes more severe over time. The pain may be constant or intermitten
```

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for ad
response(user_input_3) #Complete the code to pass the user input
```

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has :
response(user_input_4) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nA person who has sustained a physical injury to brain tissue, als
o known as a traumatic brain injury (TBI), may require various treatme
nts depending on the severity and location of the injury. Here are som
e common treatments recommended for TBIs:\n\n1. Emergency care: The fi
rst priority is to ensure the person's airway is clear, they are breat
hing, and their heart is beating normally. In severe cases, emergency
surgery may be required to remove hematomas or other obstructions.\n2.
```

Query 5: What are the necessary precautions and treatment steps

- ✓ for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"
response(user_input_5) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat,
```

- ✓ Question Answering using LLM with Prompt Engineering

```
system_prompt = "You are an experienced medical doctor. Provide accurate and concise answers to medical questions."
user_input = "What is the protocol for managing sepsis in a critical care unit?"
response(system_prompt + user_input)
```

Double-click (or enter) to edit

- ✓ Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = system_prompt+"\n" + "What is the protocol for managing sepsis in a critical care unit?"
response(user_input)
```

```
Llama.generate: prefix-match hit
'\n\nSepsis is a life-threatening condition caused by a dysregulated response to an infection. In a critical care unit, managing sepsis involves prompt recognition, early goal-directed therapy (EGDT), and ongoing supportive care. Here are the key steps:\n\n1. Recognition: Look for signs of infection, such as fever, chills, or localized inflammation, along with systemic symptoms like tachycardia, tachypnea, altered mental
```

- Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input = system_prompt + "\n" + "What are the common symptoms for  
#Complete the code to pass the query #2  
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\nAppendicitis is a common inflammatory condition of the appendix, a  
small tube-shaped organ located in the lower right side of the abdom  
n. The symptoms of appendicitis typically include:\n1. Abdominal pain:  
The pain usually starts as a mild discomfort around the navel that may  
come and go. However, it soon localizes to the lower right side of the  
abdomen and becomes constant and severe. The pain may be worsened by w
```

- Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input = system_prompt + "\n" + "What are the effective treatment:  
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit  
'\n\nSudden patchy hair loss, also known as alopecia areata, is an aut  
oimmune disorder that results in the sudden loss of hair in small, rou  
nd patches on the scalp or other areas of the body. The exact cause of  
alopecia areata is not known, but it's believed to be related to a pro  
blem with the immune system.\n\nThere are several treatments for addre  
ssing sudden patchy hair loss:\n\n1. Corticosteroids: These medication
```

- Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input = system_prompt + "\n" + "What treatments are recommended"
#Complete the code to pass the query #4
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nA person with a brain injury may require various treatments depending on the severity and location of the injury. Here are some common treatments:\n\n1. Emergency care: For severe brain injuries, emergency care is essential to prevent further damage and ensure the patient's safety. This may include surgery to remove hematomas or other obstructions, intubation and mechanical ventilation, and administration of medications to manage increased intracranial pressure or seizures.\n2. Rehabilitation
```

Query 5: What are the necessary precautions and treatment steps

- ✓ for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input = system_prompt + "\n" + "What are the necessary precautions"
#Complete the code to pass the query #5
response(user_input) #Complete the code to pass the user input
```

```
Llama.generate: prefix-match hit
'\n\nA fractured leg, also known as a broken leg, is a common injury that can occur during hiking due to falls, tripping over rocks or roots, or other traumatic events. The necessary precautions and treatment steps for a person with a fractured leg are as follows:\n\n1. Assess the severity of the injury: If you suspect a fracture, do not move the person unless it is necessary to prevent further harm or danger. Check for
```

- ✓ Data Preparation for RAG

```
import sys
import subprocess

print("Python:", sys.version)
print("Executable:", sys.executable)

subprocess.run(["pip", "list"])
```

```
Python: 3.12.13 (main, Mar 4 2026, 09:23:07) [GCC 11.4.0]
Executable: /usr/bin/python3
CompletedProcess(args=['pip', 'list'], returncode=0)
```

```
#!pip list | grep -E "langchain|chroma|sentence|huggingface|pymupdf|t"
```

```
#!pip install langchain==1.3.9 langchain-community==0.4.2 langchain-t"
```

```
import numpy as np
print(np.__version__)
```

```
2.2.6
```

```
#!pip install --force-reinstall --no-cache-dir numpy==1.26.4 scipy==1
```

```
Collecting numpy==1.26.4
  Downloading numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinu
    61.0/61.0 kB 98.1 MB/s
Collecting scipy==1.11.4
  Downloading scipy-1.11.4-cp312-cp312-manylinux_2_17_x86_64.manylinu
    60.4/60.4 kB 249.0 MB/s
Collecting scikit-learn==1.3.2
  Downloading scikit_learn-1.3.2-cp312-cp312-manylinux_2_17_x86_64.ma
Collecting joblib>=1.1.1 (from scikit-learn==1.3.2)
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting threadpoolctl>=2.0.0 (from scikit-learn==1.3.2)
  Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Downloading numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2
    18.0/18.0 MB 96.0 MB/s et
Downloading scipy-1.11.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2
    35.8/35.8 MB 102.6 MB/s e
Downloading scikit_learn-1.3.2-cp312-cp312-manylinux_2_17_x86_64.many
    10.8/10.8 MB 134.5 MB/s e
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
    309.1/309.1 kB 292.2 MB/s
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
```

```

Installing collected packages: threadpoolctl, numpy, joblib, scipy, s
Attempting uninstall: threadpoolctl
  Found existing installation: threadpoolctl 3.6.0
  Uninstalling threadpoolctl-3.6.0:
    Successfully uninstalled threadpoolctl-3.6.0
Attempting uninstall: numpy
  Found existing installation: numpy 1.26.4
  Uninstalling numpy-1.26.4:
    Successfully uninstalled numpy-1.26.4
Attempting uninstall: joblib
  Found existing installation: joblib 1.5.3
  Uninstalling joblib-1.5.3:
    Successfully uninstalled joblib-1.5.3
Attempting uninstall: scipy
  Found existing installation: scipy 1.11.4
  Uninstalling scipy-1.11.4:
    Successfully uninstalled scipy-1.11.4
Attempting uninstall: scikit-learn
  Found existing installation: scikit-learn 1.3.2
  Uninstalling scikit-learn-1.3.2:
    Successfully uninstalled scikit-learn-1.3.2

```

```

ERROR: pip's dependency resolver does not currently take into account
inequality 1.1.2 requires scipy>=1.12, but you have scipy 1.11.4 whic
opencv-contrib-python 4.13.0.92 requires numpy>=2; python_version >=
jaxlib 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is
jaxlib 0.7.2 requires scipy>=1.13, but you have scipy 1.11.4 which is
cuml-cu12 26.2.0 requires scikit-learn>=1.5, but you have scikit-lear
cuml-cu12 26.2.0 requires scipy>=1.13.0, but you have scipy 1.11.4 wh
hdbscan 0.8.44 requires scikit-learn>=1.6, but you have scikit-learn
pointpats 2.5.5 requires scipy>=1.12, but you have scipy 1.11.4 which
libpysal 4.14.1 requires scikit-learn>=1.4.0, but you have scikit-lea
libpysal 4.14.1 requires scipy>=1.12.0, but you have scipy 1.11.4 whi
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is
tobler 0.14.0 requires scipy>=1.13, but you have scipy 1.11.4 which i
opencv-python-headless 4.13.0.92 requires numpy>=2; python_version >=
imbalanced-learn 0.14.2 requires scikit-learn<2,>=1.4.2, but you have
esda 2.9.0 requires scikit-learn>=1.4, but you have scikit-learn 1.3.
esda 2.9.0 requires scipy>=1.12, but you have scipy 1.11.4 which is i
rasterio 1.5.0 requires numpy>=2, but you have numpy 1.26.4 which is
tsfresh 0.21.2 requires scipy>=1.14.0; python_version >= "3.10", but
shap 0.52.0 requires numpy>=2, but you have numpy 1.26.4 which is inc
pytensor 2.38.3 requires numpy>=2.0, but you have numpy 1.26.4 which
tiff file 2026.4.11 requires numpy>=2.0, but you have numpy 1.26.4 whi
spopt 0.7.0 requires scikit-learn>=1.4.0, but you have scikit-learn 1
spopt 0.7.0 requires scipy>=1.12.0, but you have scipy 1.11.4 which i
access 1.1.10.post3 requires scipy>=1.14.1, but you have scipy 1.11.4
cupy-cuda12x 14.0.1 requires numpy<2.6,>=2.0, but you have numpy 1.26
mapclassify 2.10.0 requires scikit-learn>=1.4, but you have scikit-le
mapclassify 2.10.0 requires scipy>=1.12, but you have scipy 1.11.4 wh
umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-lea

```

opencv-python 4.13.0.92 requires numpy>=2; python\_version >= "3.9", but you have numpy 1.26.4  
xarray-einstats 0.10.0 requires numpy>=2.0, but you have numpy 1.26.4  
xarray-einstats 0.10.0 requires scipy>=1.13, but you have scipy 1.11.  
jax 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is in  
jax 0.7.2 requires scipy>=1.13, but you have scipy 1.11.4 which is in  
Successfully installed joblib-1.5.3 numpy-1.26.4 scikit-learn-1.3.2 s

**WARNING: The following packages were previously imported in this runtime:**

[numpy]

**You must restart the runtime in order to use newly installed versions.**

RESTART SESSION

```
import numpy as np
import scipy
import sklearn
```

```
print("numpy:", np.__version__)
print("scipy:", scipy.__version__)
print("sklearn:", sklearn.__version__)
```

```
numpy: 1.26.4
scipy: 1.11.4
sklearn: 1.3.2
```

```
!pip install -q langchain-huggingface==0.1.2
```

```

461.3/461.3 kB 30.9 MB/s eta 0s
66.5/66.5 kB 7.0 MB/s eta 0s
ERROR: pip's dependency resolver does not currently take into account
langchain-text-splitters 1.1.2 requires langchain-core<2.0.0,>=1.2.31,
langchain-community 0.4.2 requires langchain-core<2.0.0,>=1.4.0, but you
langchain-classic 1.0.8 requires langchain-core<2.0.0,>=1.4.4, but you
cuml-cu12 26.2.0 requires scikit-learn>=1.5, but you have scikit-learn
cuml-cu12 26.2.0 requires scipy>=1.13.0, but you have scipy 1.11.4 which
pointpats 2.5.5 requires scipy>=1.12, but you have scipy 1.11.4 which
libpysal 4.14.1 requires scikit-learn>=1.4.0, but you have scikit-learn
libpysal 4.14.1 requires scipy>=1.12.0, but you have scipy 1.11.4 which
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is
tobler 0.14.0 requires scipy>=1.13, but you have scipy 1.11.4 which is
langchain 1.3.9 requires langchain-core<2.0.0,>=1.4.6, but you have lan
langgraph 1.2.5 requires langchain-core<2,>=1.4.7, but you have langcha
esda 2.9.0 requires scikit-learn>=1.4, but you have scikit-learn 1.3.2
esda 2.9.0 requires scipy>=1.12, but you have scipy 1.11.4 which is in
langgraph-prebuilt 1.1.0 requires langchain-core>=1.3.1, but you have
tsfresh 0.21.2 requires scipy>=1.14.0; python_version >= "3.10", but yo
shap 0.52.0 requires numpy>=2, but you have numpy 1.26.4 which is incor
spopt 0.7.0 requires scikit-learn>=1.4.0, but you have scikit-learn 1.
spopt 0.7.0 requires scipy>=1.12.0, but you have scipy 1.11.4 which is
langgraph-sdk 0.4.2 requires langchain-core<2,>=1.4.0, but you have lan
access 1.1.10.post3 requires scipy>=1.14.1, but you have scipy 1.11.4 v
google-adk 1.29.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you l
google-adk 1.29.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you l
xarray-einstats 0.10.0 requires numpy>=2.0, but you have numpy 1.26.4 v
xarray-einstats 0.10.0 requires scipy>=1.13, but you have scipy 1.11.4

```

```
!pip show langchain-huggingface
```

```

Name: langchain-huggingface
Version: 0.1.2
Summary: An integration package connecting Hugging Face and LangChain
Home-page: https://github.com/langchain-ai/langchain
Author:
Author-email:
License: MIT
Location: /usr/local/lib/python3.12/dist-packages
Requires: huggingface-hub, langchain-core, sentence-transformers, token
Required-by:

```

```
from langchain_huggingface import HuggingFaceEmbeddings

embedding_model = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your setti
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to ac
```

```
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenti
```

```
modules.json: 100% 349/349 [00:00<00:00, 21.6kB/s]
```

```
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 10.5kB/s]
```

```
README.md: 100% 10.5k/10.5k [00:00<00:00, 1.11MB/s]
```

```
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 5.80kB/s]
```

```
config.json: 100% 612/612 [00:00<00:00, 70.1kB/s]
```

```
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 50.1MB/s]
```

```
Loading weights: 100% 103/103 [00:00<00:00, 3688.58it/s]
```

```
tokenizer_config.json: 100% 350/350 [00:00<00:00, 32.0kB/s]
```

```
vocab.txt: 100% 232k/232k [00:00<00:00, 15.1MB/s]
```

```
tokenizer.json: 100% 466k/466k [00:00<00:00, 20.3MB/s]
```

```
special_tokens_map.json: 100% 112/112 [00:00<00:00, 9.79kB/s]
```

```
config.json: 100% 190/190 [00:00<00:00, 20.7kB/s]
```

```
embedding_1 = embedding_model.embed_query(document_chunks[0].page_content)
print(len(embedding_1))
```

```
-----
NameError                                Traceback (most recent call
last)
/tmp/ipykernel_11048/1028649522.py in <cell line: 0>()
----> 1 embedding_1 =
embedding_model.embed_query(document_chunks[0].page_content)
      2 print(len(embedding_1))
```

Next steps: [Explain error](#)

```
import json, os
import tiktoken
import pandas as pd
```

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.vectorstores import Chroma
from langchain_huggingface import HuggingFaceEmbeddings
```

```
# Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd
```

```
# Libraries for Loading Data, Chunking, Embedding, and Vector Database
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
#from langchain_community.embeddings import SentenceTransformerEmbedder
from langchain_community.vectorstores import Chroma
```

Start coding or [generate](#) with AI.

```
# Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd

# Libraries for Loading Data, Chunking, Embedding, and Vector Database
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
#from langchain_community.embeddings import SentenceTransformerEmbedder
from langchain_community.vectorstores import Chroma
```

```
#Libraries for processing dataframes,text
import json,os
import tiktoken
import pandas as pd

#Libraries for Loading Data, Chunking, Embedding, and Vector Database
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
#from langchain_community.embeddings import SentenceTransformerEmbedder
from langchain_community.vectorstores import Chroma
```

```
import json, os
import tiktoken
import pandas as pd

from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.vectorstores import Chroma
from langchain_huggingface import HuggingFaceEmbeddings
```

```
#!pip uninstall -y numpy
#!pip install -q --no-cache-dir numpy==2.2.6
```

## ✓ Loading the Data

```
from google.colab import drive
drive.mount('/content/drive')

manual_pdf_path = "/content/drive/MyDrive/Project_NLP w:Gen AI_Medica

pdf_loader = PyMuPDFLoader(manual_pdf_path)
manual = pdf_loader.load()
```

Mounted at /content/drive

```
# ive.mount('/content/drive')# uncomment and run the below code snipp
# from google.colab import drive
# dr
```

```
manual_pdf_path = "/content/drive/MyDrive/Project_NLP w:Gen AI_Medica
```

```
manual_pdf_path = "/content/drive/MyDrive/Project_NLP w:Gen AI_Medica

pdf_loader = PyMuPDFLoader(manual_pdf_path)
manual = pdf_loader.load()
```

Start coding or [generate](#) with AI.

```
# manual_pdf_path = "____" #Complete the code to define the file name
```

```
import os

manual_pdf_path = "/content/drive/MyDrive/Project_NLP w:Gen AI_Medica

print(os.path.exists(manual_pdf_path))
```

True

```
# pdf_loader = PyMuPDFLoader(manual_pdf_path)

# manual = pdf_loader.load()
```

```
# pdf_loader = PyMuPDFLoader(manual_pdf_path)
```

```

from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings import SentenceTransformerEmbedder
from langchain_community.vectorstores import Chroma

```

## ✓ Data Overview

## ✓ Checking the first 5 pages

```

for i in range(5):
    print(f"Page Number : {i+1}",end="\n")
    print(manual[i].page_content,end="\n")

```

```

Chapter 12. Esophageal & Swallowing Disorders .....
183
Chapter 13. Gastritis & Peptic Ulcer Disease .....
196
Chapter 14. Bezoars & Foreign Bodies .....
199
Chapter 15. Pancreatitis .....
206
Chapter 16. Gastroenteritis .....
213
Chapter 17. Malabsorption Syndromes .....
225
Chapter 18. Irritable Bowel Syndrome .....
229
Chapter 19. Inflammatory Bowel Disease .....
241
Chapter 20. Diverticular Disease .....
246
Chapter 21. Anorectal Disorders .....
254
Chapter 22. Tumors of the GI Tract .....
275
3 – Hepatic & Biliary Disorders .....
275
Chapter 23. Approach to the Patient With Liver Disease .....
294
Chapter 24. Testing for Hepatic & Biliary Disorders .....
305
Chapter 25. Drugs & the Liver .....
308
Chapter 26. Alcoholic Liver Disease .....
314

```

|                                                              |     |
|--------------------------------------------------------------|-----|
| Chapter 27. Fibrosis & Cirrhosis .....                       | 322 |
| Chapter 28. Hepatitis .....                                  | 333 |
| Chapter 29. Vascular Disorders of the Liver .....            | 341 |
| Chapter 30. Liver Masses & Granulomas .....                  | 348 |
| Chapter 31. Gallbladder & Bile Duct Disorders .....          | 362 |
| 4 – Musculoskeletal & Connective Tissue Disorders .....      | 362 |
| Chapter 32. Approach to the Patient With Joint Disease ..... | 373 |
| Chapter 33. Autoimmune Rheumatic Disorders .....             | 391 |
| Chapter 34. Vasculitis .....                                 | 416 |
| Chapter 35. Joint Disorders .....                            | 435 |
| Chapter 36. Crystal-Induced Arthritides .....                | 443 |
| Chapter 37. Osteoporosis .....                               | 448 |
| Chapter 38. Paget's Disease of Bone .....                    | 451 |
| Chapter 39. Osteonecrosis .....                              |     |

## ✓ Checking the number of pages

```
len(manual)
```

```
4114
```

## ✓ Data Chunking

```
text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name='cl100k_base',
    chunk_size=1000,
    chunk_overlap=200
)
```

```
document_chunks = pdf_loader.load_and_split(text_splitter)
```

```
len(document_chunks)
```

```
4675
```

```
document_chunks[0].page_content
```

```
'sarahrwolberg@gmail.com\n6KTHCMR782\nThis file is meant for personal
use by sarahrwolberg@gmail.com only.\nSharing or publishing the conten
ts in part or full is liable for legal action '
```

```
document_chunks[2].page_content
```

```
'Table of Contents\n1\nFront
.....
\n1\nCover
.....
\n2\nFront Matter
.....
\n53\n1 - Nutritional Disorders
.....
\n53\nChapter 1. Nutrition: General Considerations
.....
...'
```

```
document_chunks[3].page_content
```

```
'491\nChapter 44. Foot & Ankle Disorders
.....
\n502\nChapter 45. Tumors of Bones & Joints
.....
\n510\n5 - Ear, Nose, Throat & Dental Disorders
.....
\n510\nChapter 46. Approach to the Patient With Ear Problems
.....
\n523\nChapter 47. Hearing Loss
.....
\n535\nChapter 48. Inner Ear Disorders
.....
...'
```

As expected, there are some overlaps

## ✓ Embedding

```
#!pip install -U sentence-transformers
```

```
#!pip install -U langchain-huggingface
```

```
#!pip install -q -U langchain-huggingface
```

```
!#pip show langchain-huggingface
```

```
#!pip uninstall -y sentence-transformers langchain-huggingface
!pip install -q sentence-transformers==3.0.1 langchain-huggingface==0.1
```

```
from langchain_huggingface import HuggingFaceEmbeddings

embedding_model = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
```

Loading weights: 100% 103/103 [00:00<00:00, 3437.49it/s]

```
import transformers
import sentence_transformers

print("Transformers:", transformers.__version__)
print("Sentence Transformers:", sentence_transformers.__version__)
```

Transformers: 5.12.0  
Sentence Transformers: 5.5.1

```
from langchain_huggingface import HuggingFaceEmbeddings
embedding_model = HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-v2")
```

Loading weights: 100% 103/103 [00:00<00:00, 1747.70it/s]

```
from langchain_huggingface import HuggingFaceEmbeddings
embedding_model = HuggingFaceEmbeddings(model_name="sentence-transformer
```

Loading weights: 100%

103/103 [00:00<00:00, 4542.49it/s]

```
embedding_1 = embedding_model.embed_query(document_chunks[0].page_content)
embedding_2 = embedding_model.embed_query(document_chunks[1].page_content)
```

```
print("Dimension of the embedding vector ", len(embedding_1))
len(embedding_1)==len(embedding_2)
```

Dimension of the embedding vector 384  
True

```
embedding_1, embedding_2
```

```
0.00990404379000247,
-0.013455205596983433,
0.08267802745103836,
0.015714911743998528,
-0.05515673756599426,
-0.004949823021888733,
0.11951562762260437,
-0.007674371358007193,
-0.09202570468187332,
0.01300744153559208,
-0.07599160075187683,
0.003223179141059518,
-0.0005568215274251997,
0.08737407624721527,
0.01239523384720087,
-0.00817912444472313,
0.025717411190271378,
-0.03492218255996704,
0.004231343977153301,
-0.056925371289253235,
0.03121616691350937,
0.061887580901384354,
0.032487668097019196,
-0.06689317524433136,
0.03301270678639412,
-0.016769006848335266,
-0.027753081172704697,
-0.07893133908510208,
0.09871811419725418,
-0.05034981667995453,
```

```
-0.047350212931632996,  
0.06373202800750732,  
0.09545337408781052,  
-0.025847120210528374,  
0.031696148216724396,  
-0.03266061469912529,  
-0.03464510664343834,  
0.0011421822709962726,  
-0.10882682353258133,  
-0.15245576202869415,  
0.08895636349916458,  
8.284714472598582e-33,  
-0.013746914453804493,  
0.015155482105910778,  
0.08626843243837357,  
0.036270737648010254,  
0.014086320996284485,  
-0.03352930024266243,  
-0.01592225581407547,  
0.0013393198605626822,  
-0.1400613933801651,  
-0.0799468606710434,  
-0.026157651096582413,  
-0.01706414297223091,  
0.021833952516317368,  
0.024200724437832832,  
-0.10236099362373352,  
0.05255921930074692,  
0.011665739119052887,  
0.03520475700497627.
```

## ✓ Vector Database

```
out_dir = 'medical_db'
```

```
if not os.path.exists(out_dir):  
    os.makedirs(out_dir)
```

```
vectorstore = Chroma.from_documents(  
    documents=document_chunks,      # pass the document chunks  
    embedding=embedding_model,      # pass the embedding model  
    persist_directory=out_dir  
)
```

```
vectorstore = Chroma(persist_directory=out_dir,embedding_function=emb
```

```
/tmp/ipykernel_11048/2756559696.py:1: LangChainDeprecationWarning: The
vectorstore = Chroma(persist_directory=out_dir,embedding_function=eml
```

```
vectorstore.embeddings
```

```
HuggingFaceEmbeddings(model_name='sentence-transformers/all-MiniLM-L6-
v2', cache_folder=None, model_kwargs={}, encode_kwargs={},
multi_process=False, show_progress=False)
```

```
vectorstore.similarity_search("What is the protocol for managing seps:
```

adults or 250 to 500 mg/m<sup>2</sup> body surface area IV q 8 h for children is\nrecommended for varicella lung infections. Some patients with viral pneumonia, especially those with\ninfluenza, develop superimposed bacterial infections and require antibiotics directed against S.\npneumoniae, H. influenzae, and Staphylococcus aureus.\nWith empiric treatment, 90% of patients with bacterial pneumonia improve. Improvement is manifested by\ndeclined cough and dyspnea, defervescence, relief of chest pain, and decline in WBC count. Failure to\nimprove should trigger suspicion of an unusual organism, resistance to the antimicrobial used for\ntreatment, empyema, coinfection or superinfection with a 2nd infectious agent, an obstructive\nendobronchial lesion, immunosuppression, metastatic focus of infection with reseeding (in the case of\npneumococcal infection), or nonadherence to treatment (in the case of outpatients). If none of these can\nbe proved, treatment failure is likely due to inadequate host defenses.\nMost viral pneumonias resolve without specific treatment.\nChest physical therapy can be used to treat pneumonia; however, there is no clear evidence for its\nefficacy. Follow-up x-rays should be obtained 6 wk after treatment in patients > 35; persistence of an\ninfiltrate at ≥ 6 wk raises suspicions of an underlying, possibly malignant endobronchial lesion or of TB.\nPrevention\nSome forms of community-acquired pneumonia are preventable with pneumococcal conjugate vaccine\n(for patients < 2 yr), H. influenzae type b (Hib) vaccine (for patients < 2 yr), pneumococcal pneumonia\nvaccine (for patients at high risk, such as those with underlying heart, lung, or immune system disorders),\nvaricella vaccine (for patients < 18 mo and a later booster vaccine), and influenza vaccine (for patients\nage ≥ 65 and those at high risk—see\nTable 131-2 on p. 1174 and see\nTable 268-10 on p. 2718). Oseltamivir 75 mg po once/day or zanamivir 10 mg once/day can be given for 2\nwk to prevent influenza (although resistance has recently been described for oseltamivir) for household\ncontacts of patients with influenza and to high-risk patients not vaccinated against influenza during\ninfluenza

```

epidemics. Pneumococcal pneumonia vaccination is recommended for all
patients ≥ 65 (see p.\n1177).\nHospital-Acquired Pneumonia\nThe
Merck Manual of Diagnosis & Therapy, 19th Edition\nChapter 196.
Pneumonia\n2085\nnsarahrwolberg@gmail.com'),
Document(metadata={'file_path': '/content/drive/MyDrive/Project_NLP
w:Gen AI_Medical Assistant/medical_diagnosis_manual.pdf',
'total_pages': 4114, 'subject': '', 'keywords': '', 'title': 'The
Merck Manual of Diagnosis & Therapy, 19th Edition', 'creationdate':
'2012-06-15T05:44:40+00:00', 'creator': 'Atop CHM to PDF Converter',
'author': '', 'page': 2995, 'producer': 'pdf-lib
(https://github.com/Hopding/pdf-lib)', 'source':
'/content/drive/MyDrive/Project_NLP w:Gen AI_Medical
Assistant/medical_diagnosis_manual.pdf', 'trapped': '', 'moddate':
'2026-06-27T16:38:14+00:00', 'format': 'PDF 1.7', 'modDate':
'D:20260627163814Z', 'creationDate': 'D:20120615054440Z'},
page_content='can approximate bone marrow NSP levels. I:T ratios of
> 0.80 correlate with NSP depletion and death;\nsuch a ratio may
identify neonates who might benefit from granulocyte
transfusion.\nTreatment\n• Antibiotic therapy\n• Supportive
therapy\nBecause sepsis may manifest with non-specific clinical
signs and its effects may be devastating, rapid\nempiric antibiotic
therapy is recommended (see p. 1182); drugs are later adjusted
according to\nsensitivities and the site of infection. If bacterial
cultures show no growth by 48 h (although some\npathogens may
require 72 h) and the neonate appears well, antibiotics are

```

## ▼ Retriever

```

retriever = vectorstore.as_retriever(
    search_type='similarity',
    search_kwargs={'k': 10} #Complete the code to pass an appropriate
)

```

```

rel_docs = retriever.invoke("What is the protocol for managing sepsis
rel_docs

```

```

[Document(metadata={'keywords': '', 'modDate': 'D:20260627163814Z',
'creationDate': 'D:20120615054440Z', 'source':
'/content/drive/MyDrive/Project_NLP w:Gen AI_Medical
Assistant/medical_diagnosis_manual.pdf', 'trapped': '', 'author':
'', 'creator': 'Atop CHM to PDF Converter', 'total_pages': 4114,
'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'title':
'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'subject':
'', 'page': 2400, 'creationdate': '2012-06-15T05:44:40+00:00',
'moddate': '2026-06-27T16:38:14+00:00', 'file_path':
'/content/drive/MyDrive/Project_NLP w:Gen AI_Medical

```

Assistant/medical\_diagnosis\_manual.pdf', 'format': 'PDF 1.7'}, page\_content="16 - Critical Care Medicine\nChapter 222. Approach to the Critically Ill Patient\nIntroduction\nCritical care medicine specializes in caring for the most seriously ill patients. These patients are best\ntreated in an ICU staffed by experienced personnel. Some hospitals maintain separate units for special\npopulations (eg, cardiac, surgical, neurologic, pediatric, or neonatal patients). ICUs have a high\nnurse:patient ratio to provide the necessary high intensity of service, including treatment and monitoring\nof physiologic parameters.\nSupportive care for the ICU patient includes provision of adequate nutrition (see p. 21) and prevention of\ninfection, stress ulcers and gastritis (see p. 131), and pulmonary embolism (see p. 1920). Because 15 to\n25% of patients admitted to ICUs die there, physicians should know how to minimize suffering and help\nndying patients maintain dignity (see p. 3480).\nPatient Monitoring and Testing\nSome monitoring is manual (ie, by direct observation and physical examination) and intermittent, with the\nfrequency depending on the patient's illness. This monitoring usually includes measurement of vital signs\n(temperature, BP, pulse, and respiration rate), quantification of all fluid intake and output, and often daily\nweight. BP may be recorded by an automated sphygmomanometer; a transcutaneous sensor for pulse\noximetry is used as well.\nOther monitoring is ongoing and continuous, provided by complex devices that require special training\nand experience to operate. Most such devices generate an alarm if certain physiologic parameters are\nexceeded. Every ICU should strictly follow protocols for investigating alarms.\nBlood Tests\nAlthough frequent blood draws can destroy veins, cause pain, and lead to anemia, ICU patients typically\nhave routine daily blood tests to help detect problems early. Generally, patients need a daily set of\nelectrolytes and a CBC. Patients with arrhythmias should also have Mg, phosphate, and Ca levels\nmeasured. Patients receiving TPN need weekly liver enzymes and coagulation profiles. Other tests (eg,\nblood culture for fever, CBC after a bleeding episode) are done as needed.\nPoint-of-care testing uses miniaturized, highly automated devices to do certain blood tests at the patient's\nbedside or unit (particularly ICU, emergency department, and operating room). Commonly available tests\ninclude blood chemistries, glucose, ABGs, CBC, cardiac markers, and coagulation tests. Many are done\nin < 2 min and require < 0.5 mL blood.\nCardiac Monitoring\nMost critical care patients have cardiac activity monitored by a 3-lead system; signals are usually sent to\na central monitoring station by a small radio transmitter worn by the patient. Automated systems generate\nalarms for abnormal rates and rhythms and store abnormal tracings for subsequent review.\nSome specialized cardiac monitors track advanced parameters associated with coronary ischemia,\nalthough their clinical benefit is unclear. These parameters include continuous ST-segment monitoring\nand heart rate variability. Loss of normal heart-

```

from huggingface_hub import hf_hub_download
from llama_cpp import Llama

model_name_or_path = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"
model_basename = "mistral-7b-instruct-v0.2.Q6_K.gguf"

model_path = hf_hub_download(
    repo_id=model_name_or_path,
    filename=model_basename
)

llm = Llama(
    model_path=model_path,
    n_ctx=8192,
    n_gpu_layers=38,
    n_batch=512
)

```

AVX = 1 | AVX2 = 1 | AVX512 = 0 | AVX512\_VBMI = 0 | AVX512\_VNNI = 0 |

```

model_output = llm(
    "What is the protocol for managing sepsis in a critical care unit"
    max_tokens=256,
    temperature=0
)

```

```
model_output["choices"][0]["text"]
```

'\n\nSepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit:\n\n1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.\n2. Resuscitation: Provide adequate fluid resuscitation to maintain adequate tissue perfusion. The goal is to achieve a mean arterial pressure (MAP) of at least 65 mmHg and a central venous oxygen saturation (ScvO2) of greater than 70%.

```
# model_output = llm(
#     "What is the protocol for managing sepsis in a critical care u
#     max_tokens=256, #Complete the code to pass the maximum number
#     temperature=0, #Complete the code to pass the temperature
# )
```

```
model_output['choices'][0]['text']
```

```
'\n\nSepsis is a life-threatening condition that can arise from an inf
ection, and it requires prompt recognition and aggressive management i
n a critical care unit. The following are general steps for managing s
epsis in a critical care unit:\n\n1. Early recognition: Recognize the
signs and symptoms of sepsis early and initiate treatment as soon as p
ossible. Sepsis can present with various clinical features, including
fever or hypothermia, tachycardia or bradycardia, altered mental statu
s, respiratory distress, and lactic acidosis.\n2. Resuscitation: Provi
de adequate fluid resuscitation to maintain adequate tissue perfusion.
The goal is to achieve a mean arterial pressure (MAP) of at least 65 m
mHg and a central venous oxygen saturation (ScvO2) of greater than 7
```

The above response is somewhat generic and is solely based on the data the model was trained on, rather than the medical manual.

Let's now provide our own context.

## ✓ System and User Prompt Template

Prompts guide the model to generate accurate responses. Here, we define two parts:

1. The system message describing the assistant's role.
2. A user message template including context and the question.

```
qna_system_message = ""
You are a knowledgeable medical assistant. Answer the user's question
"" #Complete the code to define the system message
```

```
qna_user_message_template = """
Context:
{context}

Question:
{question}

Answer:
""">#Complete the code to define the user message
```

## ✓ Response Function

```

# def generate_rag_response(user_input,k=3,max_tokens=128,temperature=
#     global qna_system_message,qna_user_message_template
#     # Retrieve relevant document chunks
#     relevant_document_chunks = retriever.get_relevant_documents(que
#     context_list = [d.page_content for d in relevant_document_chunk

#     # Combine document chunks into a single context
#     context_for_query = ". ".join(context_list)

#     user_message = qna_user_message_template.replace('{context}', co
#     user_message = user_message.replace('{question}', user_input)

#     prompt = qna_system_message + '\n' + user_message

#     # Generate the response
#     try:
#         response = llm(
#             prompt=prompt,
#             max_tokens=max_tokens,
#             temperature=temperature,
#             top_p=top_p,
#             top_k=top_k
#         )

#         # Extract and print the model's response
#         response = response['choices'][0]['text'].strip()
#     except Exception as e:
#         response = f'Sorry, I encountered the following error: \n {e}

#     return response

```

```

# def generate_rag_response(user_input, k=50, max_tokens=512, tempera
#     global qna_system_message, qna_user_message_template

#     # Retrieve relevant document chunks
#     relevant_document_chunks = retriever.invoke(user_input)

#     # Keep only the top k chunks
#     relevant_document_chunks = relevant_document_chunks[:k]

#     # Combine document chunks into a single context
#     context_for_query = "\n\n".join([d.page_content for d in releva

#     user_message = qna_user_message_template.replace("{context}", c
#     user_message = user_message.replace("{question}", user_input)

#     prompt = qna_system_message + "\n" + user_message

#     try:
#         response = llm(
#             prompt=prompt,
#             max_tokens=max_tokens,
#             temperature=temperature,
#             top_p=top_p,
#             top_k=top_k
#         )

#         response = response["choices"][0]["text"].strip()

#     except Exception as e:
#         response = f"Sorry, I encountered the following error:\n{e}"

#     return response

```

```

def generate_rag_response(user_input, k=3, max_tokens=256, temperature=0):
    global qna_system_message, qna_user_message_template

    relevant_document_chunks = retriever.invoke(user_input)
    relevant_document_chunks = relevant_document_chunks[:k]

    context_for_query = "\n\n".join([d.page_content for d in relevant_document_chunks])

    user_message = qna_user_message_template.replace("{context}", context_for_query)
    user_message = user_message.replace("{question}", user_input)

    prompt = qna_system_message + "\n" + user_message

    try:
        response = llm(
            prompt=prompt,
            max_tokens=max_tokens,
            temperature=temperature,
            top_p=top_p,
            top_k=top_k
        )
        response = response["choices"][0]["text"].strip()

    except Exception as e:
        response = f"Sorry, I encountered the following error:\n{e}"

    return response

```

```

user_input = "What is the protocol for managing sepsis in a critical care unit?"
generate_rag_response(user_input, k=3, max_tokens=256, temperature=0)

```

```

Llama.generate: prefix-match hit
'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics after cultures are obtained, adjusting antibiotics according to culture and susceptibility testing results, surgically draining any abscesses, removing internal devices that may be the source of bacteria, and providing supportive care such as fluids, antipyretics, analgesics, and oxygen for patients with hypoxemia. Early treatment with an appropriate antimicrobial regimen appears to improve s

```

```
generate_rag_response(
    "What is the protocol for managing sepsis in a critical care unit"
)
```

```
Llama.generate: prefix-match hit
'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics after cultures are obtained, adjusting antibiotics according to culture and susceptibility testing results, surgically draining any abscesses, removing internal devices that may be the source of bacteria, and providing supportive care such as fluids, antipyretics, analgesics, and oxygen for patients with hypoxemia. Early treatment with an appropriate antimicrobial regimen appears to improve s
```

## ✓ Question Answering using RAG

### ✓ Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = "What is the protocol for managing sepsis in a critical care unit?"
generate_rag_response(user_input,top_k=3)
```

```
Llama.generate: prefix-match hit
'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics after cultures are obtained, adjusting antibiotics according to culture and susceptibility testing results, surgically draining any abscesses, removing internal devices that may be the source of bacteria, and providing supportive care such as fluids, antipyretics, analgesics, and oxygen for patients with hypoxemia. Early treatment with an appropriate antimicrobial regimen appears to improve s
```

### ✓ Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can I
generate_rag_response(user_input_2,top_k=20) #Complete the code to pa
```

```
Llama.generate: prefix-match hit
'The common symptoms of appendicitis include epigastric or periumbilic
al pain followed by brief nausea, vomiting, and anorexia; after a few
hours, the pain shifts to the right lower quadrant. Pain increases wit
h cough and motion. Classic signs are right lower quadrant direct and
rebound tenderness located at McBurney's point (junction of the middle
and outer thirds of the line joining the umbilicus to the anterior sup
erior spine). Additional signs include pain felt in the right lower qu
adrant with palpation of the left lower quadrant (Rovsing sign), an in
crease in pain from passive extension of the right hip joint that stre
tches the iliopsoas muscle (psoas sign), or pain caused by passive int
ernal rotation of the flexed thigh (obturator sign). Low-grade fever (
```

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for ad
generate_rag_response(user_input_3,top_k=50) #Complete the code to pa
```

```
Llama.generate: prefix-match hit
'Sudden patchy hair loss, also known as alopecia areata, is a common a
utoimmune disorder affecting genetically susceptible individuals expos
ed to unclear environmental triggers. The most effective treatments fo
r addressing this condition include topical or intralesional corticost
eroids, topical minoxidil, topical anthralin, topical immunotherapy (d
iphencyprone or squaric acid dibutylester), or psoralen plus ultraviol
et A (PUVA). Hormonal modulators such as oral contraceptives or spiron
olactone may also be useful for female-pattern hair loss associated wi
th hyperandrogenemia. It is important to note that these treatments ai
m to stimulate hair growth and reduce inflammation, but they do not re
pair the damaged follicles. Once treatment is stopped, hair loss retur
```

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has :
generate_rag_response(user_input_4,top_k=20) #Complete the code to pa:
```

```
Llama.generate: prefix-match hit
'The manual suggests several treatments for patients with traumatic br
ain injury (TBI) that may help improve neurologic function. These incl
ude speech therapy to establish communication codes and encourage acti
ve involvement of family and friends. Additionally, there are ongoing
studies investigating the use of autologous macrophages, epidural admi
nistration of BA-210, oral administration of HP-184, and optimal timin
g of surgery for nerve regeneration. However, no treatments have been
proven effective in recovering normal neurologic function once a persi
stent vegetative state lasts for 3 months after injury and almost non
```

Query 5: What are the necessary precautions and treatment steps

- ✓ for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment step:
generate_rag_response(user_input_5,top_k=50) #Complete the code to pa:
```

```
Llama.generate: prefix-match hit
'Based on the context provided in the manual, the following steps shou
ld be taken for a person who has fractured their leg during a hiking t
rip:\n1. Assess the severity of the injury and provide first aid as ne
cessary, such as immobilizing the affected limb with a splint to preve
nt further injury and decrease pain.\n2. Treat any bleeding by applyin
g direct pressure to the wound with a clean cloth or dressing.\n3. If
there are signs of nerve or arterial damage, such as numbness, tinglin
g, or loss of pulse, seek medical attention immediately for possible s
urgical repair.\n4. For most fractures, definitive treatment involves
reduction, which may require analgesia or sedation and is usually main
tained by casting or surgical hardware.\n5. Rest, ice, compression, an
```

- ✓ Fine-tuning
- ✓ Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input = "What is the protocol for managing sepsis in a critical care unit?"
generate_rag_response(user_input,temperature=0.5)
```

```
Llama.generate: prefix-match hit
'The protocol for managing sepsis in a critical care unit includes obtaining cultures of blood and any other appropriate specimens, administering empiric antibiotics, adjusting antibiotics according to culture and susceptibility testing, surgically draining any abscesses, removing internal devices that are the suspected source of bacteria, and continuing therapy until improvement is seen or the cause of sepsis has been identified. Close monitoring for signs of hypotension, shaking chills,
```

- Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"
generate_rag_response(user_input_2,temperature=0.5) #Complete the code
```

```
Llama.generate: prefix-match hit
'The common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign), an increase in pain from passive extension of the right hip joint that stretches the iliopsoas muscle (psoas sign), or pain caused by passive internal rotation of the flexed thigh (obturator sign). Low-grade fever (
```

- Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for alopecia areata?"
generate_rag_response(user_input_3, temperature=0.5) #Complete the code
```

```
Llama.generate: prefix-match hit
'The provided context suggests that alopecia areata is a common cause of sudden patchy hair loss. The most effective treatment options for this condition include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). The choice of treatment depends on the severity and extent of hair loss. It is important to note that some treatments may have adverse effects, such as decreased libido, erectile dysfunction, hypersensitivity reactions, gynecomastia, myopathy, or a decrease in prostate-specific antigen levels in older men. Hormonal modulators such as oral
```

- Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = " What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
generate_rag_response(user_input_4, temperature=0.5) #Complete the code
```

```
Llama.generate: prefix-match hit
'The manual provides information on traumatic brain injury (TBI) and spinal trauma. For TBI, late seizures may develop weeks, months, or even years later, leading to decreased intellectual function, spastic motor impairment, gait and balance disturbances, ataxia, sensory losses, and possibly a persistent vegetative state. There is ongoing research into treatments to promote nerve regeneration, including injections of autologous macrophages, epidural administration of experimental drugs, and oral administration of HP-184 for chronic spinal cord injury. For communication, speech therapists may help establish a code using eye blinks or movements if cognitive function is intact.\n\nFor brain death, which is the loss of function of the entire cerebrum and brain stem,
```

- Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a leg fracture?"
generate_rag_response(user_input_5, temperature=0.5) #Complete the code
```

```
Llama.generate: prefix-match hit
'Based on the context provided, if a person has fractured their leg during a hiking trip, they should first receive emergency treatment for any life-threatening injuries. If there are no immediate threats to their life or limb, the leg should be immobilized using a splint to prevent further injury and decrease pain. The RICE (rest, ice, compression, elevation) method can also be applied to help minimize swelling and speed up healing. Definitive treatment for the fracture may involve reduction, which usually requires analgesia or sedation and is maintained by various surgical hardware such as pins, screws, plates, or external fixators. Immobilization with a cast is typically used for injuries that require weeks of immobilization. Patients should be instructed to
```

## ✓ Output Evaluation

Let us now use the LLM-as-a-judge method to check the quality of the RAG system on two parameters - retrieval and generation.

- We are using the same Mistral model for evaluation, so basically here the llm is rating itself on how well he has performed in the task.

```
groundedness_rater_system_message = """
You are an evaluator. Judge whether the answer is grounded in the provided context.

Rate the answer using one of these labels:
- Grounded: The answer is fully supported by the context.
- Partially Grounded: Some parts are supported, but some details are not.
- Not Grounded: The answer is mostly unsupported or contradicts the context.

Explain your rating briefly.
""" #Complete the code to define the prompt to evaluate groundedness
```

```
relevance_rater_system_message = """
You are an evaluator. Judge whether the answer directly addresses the

Rate the answer using one of these labels:
- Relevant: The answer directly and completely addresses the question
- Partially Relevant: The answer addresses part of the question but m
- Not Relevant: The answer does not address the question.

Explain your rating briefly.
""" #Complete the code to define the prompt to evaluate relevance
```

```
user_message_template = """
###Question
{question}

###Context
{context}

###Answer
{answer}
"""
```

```
# def generate_ground_relevance_response(user_input,k=3,max_tokens=128
#     global qna_system_message,qna_user_message_template
#     # Retrieve relevant document chunks
#     relevant_document_chunks = retriever.get_relevant_documents(que
#     context_list = [d.page_content for d in relevant_document_chunk
#     context_for_query = ". ".join(context_list)

#     # Combine user_prompt and system_message to create the prompt
#     prompt = f"""[INST]{qna_system_message}\n
#               {'user': {qna_user_message_template.format(context:
#               [/INST]"""

#     response = llm(
#         prompt=prompt,
#         max_tokens=max_tokens,
#         temperature=temperature,
#         top_p=top_p,
#         top_k=top_k,
#         stop=['INST'],
#         echo=False
#     )
```

```

#     answer = response["choices"][0]["text"]

#     # Combine user_prompt and system_message to create the prompt
#     groundedness_prompt = f""[INST]{groundedness_rater_system_mess;
#         {'user'}: {user_message_template.format(context=con
#         [/INST]""

#     # Combine user_prompt and system_message to create the prompt
#     relevance_prompt = f""[INST]{relevance_rater_system_message}\n
#         {'user'}: {user_message_template.format(context=con
#         [/INST]""

#     response_1 = llm(
#         prompt=groundedness_prompt,
#         max_tokens=max_tokens,
#         temperature=temperature,
#         top_p=top_p,
#         top_k=top_k,
#         stop=['INST'],
#         echo=False
#     )

#     response_2 = llm(
#         prompt=relevance_prompt,
#         max_tokens=max_tokens,
#         temperature=temperature,
#         top_p=top_p,
#         top_k=top_k,
#         stop=['INST'],
#         echo=False
#     )

#     return response_1['choices'][0]['text'], response_2['choices'][0]

```

```

def generate_ground_relevance_response(user_input, k=3, max_tokens=370)
    global qna_system_message, qna_user_message_template

    relevant_document_chunks = retriever.invoke(user_input)
    relevant_document_chunks = relevant_document_chunks[:k]

    context_list = [d.page_content for d in relevant_document_chunks]
    context_for_query = ". ".join(context_list)

```

```

prompt = f"""[INST]{qna_system_message}

user: {qna_user_message_template.format(context=context_for_query, qu
[/INST]"""

response = llm(
    prompt=prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=["INST"],
    echo=False
)

answer = response["choices"][0]["text"]

groundedness_prompt = f"""[INST]{groundedness_rater_system_message}

user: {user_message_template.format(context=context_for_query, questi
[/INST]"""

relevance_prompt = f"""[INST]{relevance_rater_system_message}

user: {user_message_template.format(context=context_for_query, questi
[/INST]"""

response_1 = llm(
    prompt=groundedness_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=["INST"],
    echo=False
)

response_2 = llm(
    prompt=relevance_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=["INST"],
    echo=False
)

```

```
)  
  
return response_1["choices"][0]["text"], response_2["choices"][0]
```

✓ Query 1: What is the protocol for managing sepsis in a critical care unit?

```
ground,rel = generate_ground_relevance_response(user_input="What is tl  
  
print(ground,end="\n\n")  
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Rating: Grounded
```

Explanation: The answer is fully supported by the context provided in

Relevant: The answer directly addresses the user's question by provid

✓ Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```

ground, rel = generate_ground_relevance_response(
    user_input="What are the common symptoms for appendicitis, and can
    max_tokens=370
) #Complete the code to pass the query #2 along with parameters if nee

print(ground,end="\n\n")
print(rel)

```

```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

```

Explanation: The user's answer is fully supported by the context provided.

Relevant: The answer directly addresses the user's question by providing

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```

ground, rel = generate_ground_relevance_response(
    user_input="What are the effective treatments or solutions for addressing
    max_tokens=370
) #Complete the code to pass the query #3 along with parameters if nee

print(ground,end="\n\n")
print(rel)

```

```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Rating: Grounded

```

Explanation: The user's answer is fully supported by the context provided.

Relevant: The answer directly addresses the user's question by providing

- Query 4: What treatments are recommended for a person who has
- ✓ sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
ground, rel = generate_ground_relevance_response(  
    user_input="What treatments are recommended for a person who has :  
    max_tokens=370  
) #Complete the code to pass the query #4 along with parameters if need  
  
print(ground,end="\n\n")  
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Rating: Grounded
```

Explanation: The context provides detailed information about the various

Relevant: The answer directly addresses the user's question by listing

- Query 5: What are the necessary precautions and treatment steps
- ✓ for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
ground, rel = generate_ground_relevance_response(  
    user_input="What are the necessary precautions and treatment steps?  
    max_tokens=370  
)  
#Complete the code to pass the query #5 along with parameters if needed  
  
print(ground,end="\n\n")  
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Rating: Grounded
```

Explanation: The context provides detailed information about the necessary precautions and treatment steps for the user's query.

Relevant: The answer directly addresses the user's question by providing the necessary precautions and treatment steps.

## ✓ Actionable Insights and Business Recommendations

- 
- 
- 

# Power Ahead

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